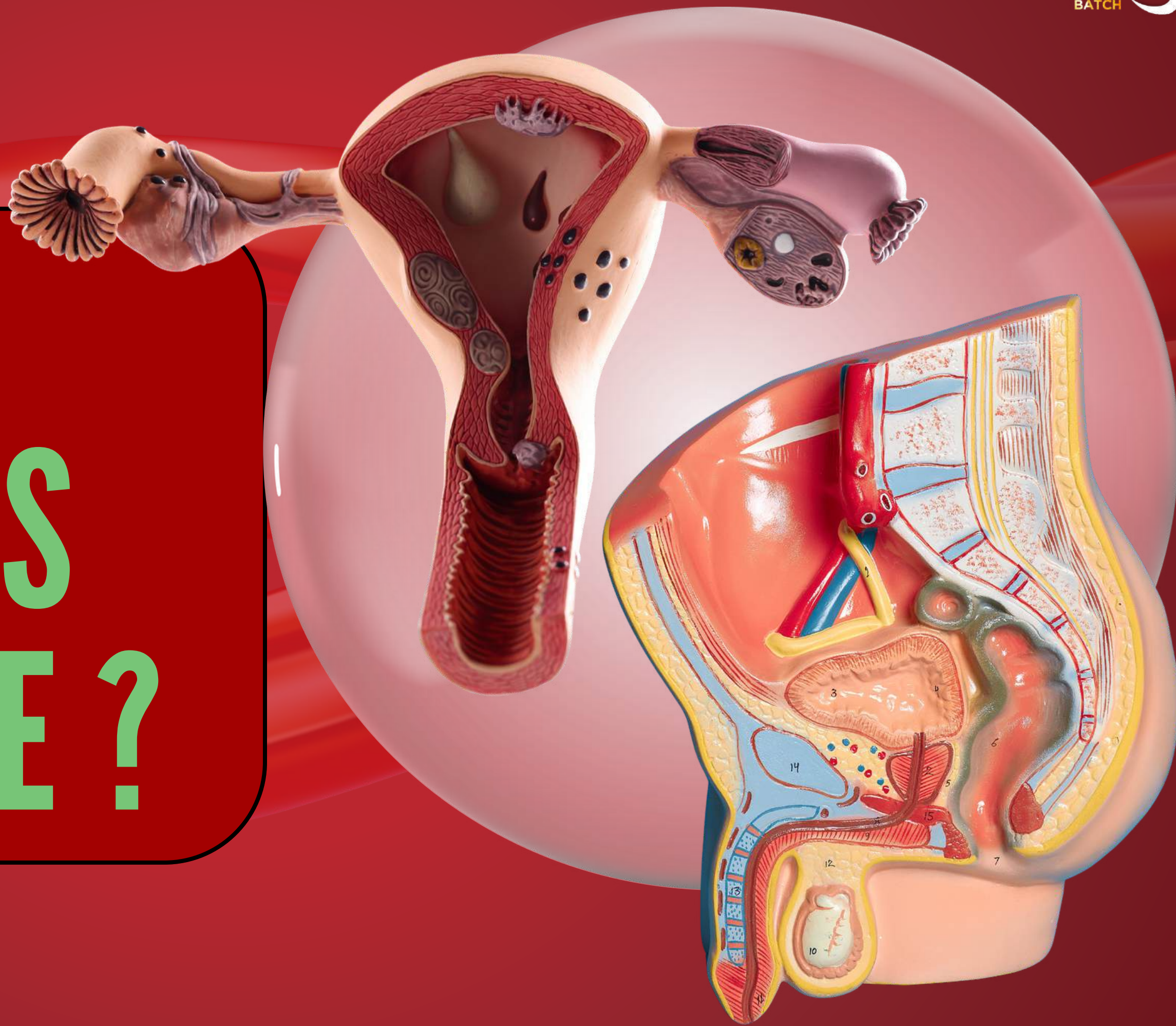


L3

CLASS X - SCIENCE

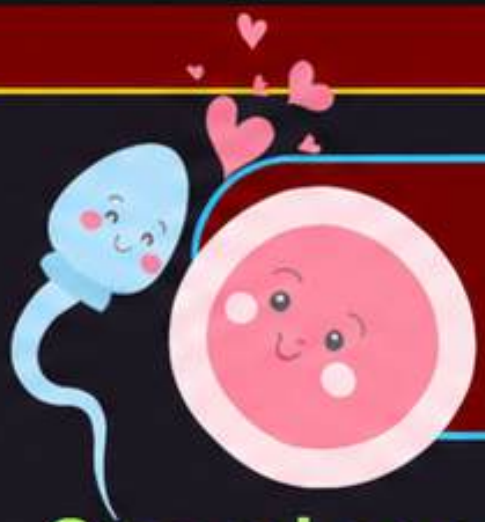
HOW DO ORGANISMS REPRODUCE ?

PRASHANT KIRAD





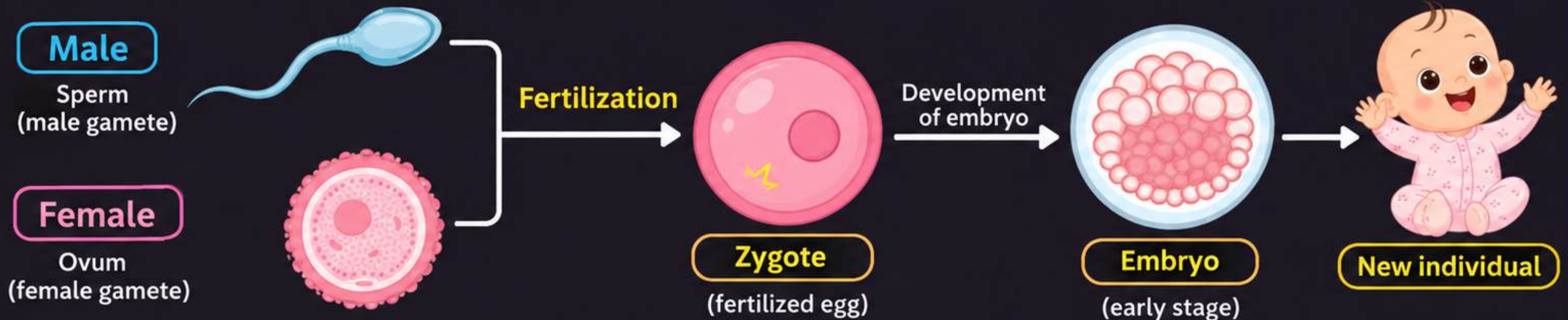
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Sexual Reproduction

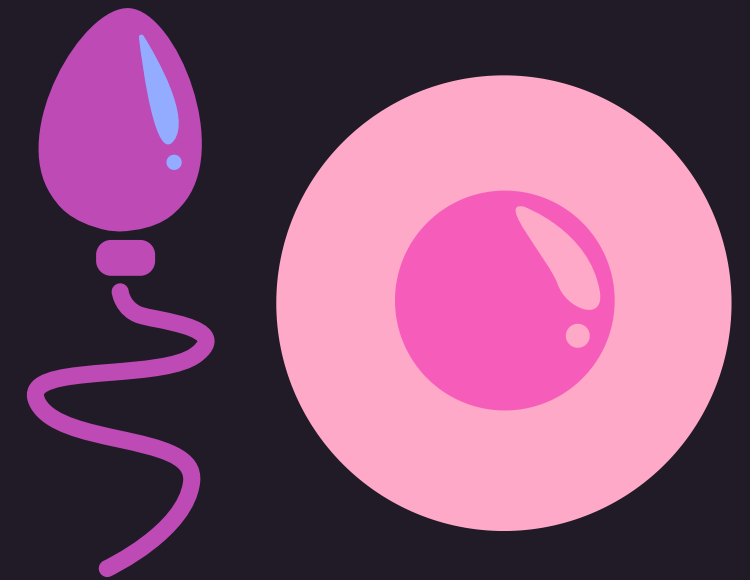
Sexual reproduction is a mode of reproduction in which two parents (male and female) are involved in producing new individuals.

- **Both sexes** are necessary.
- **Formation of gametes** occurs.
- **Fertilization** occurs.
- **Development of embryo**.



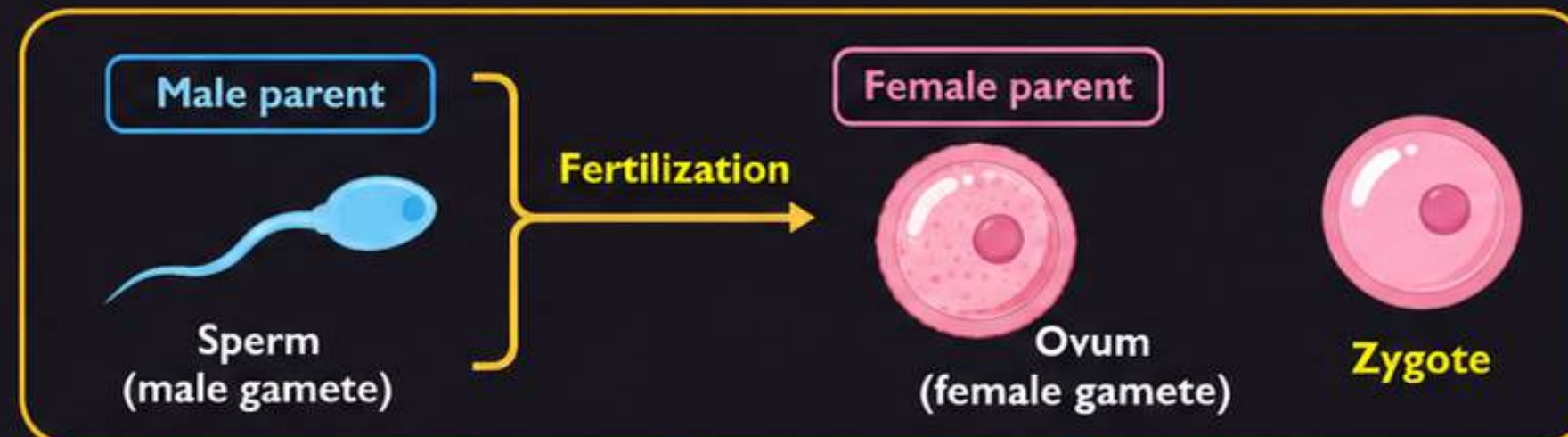
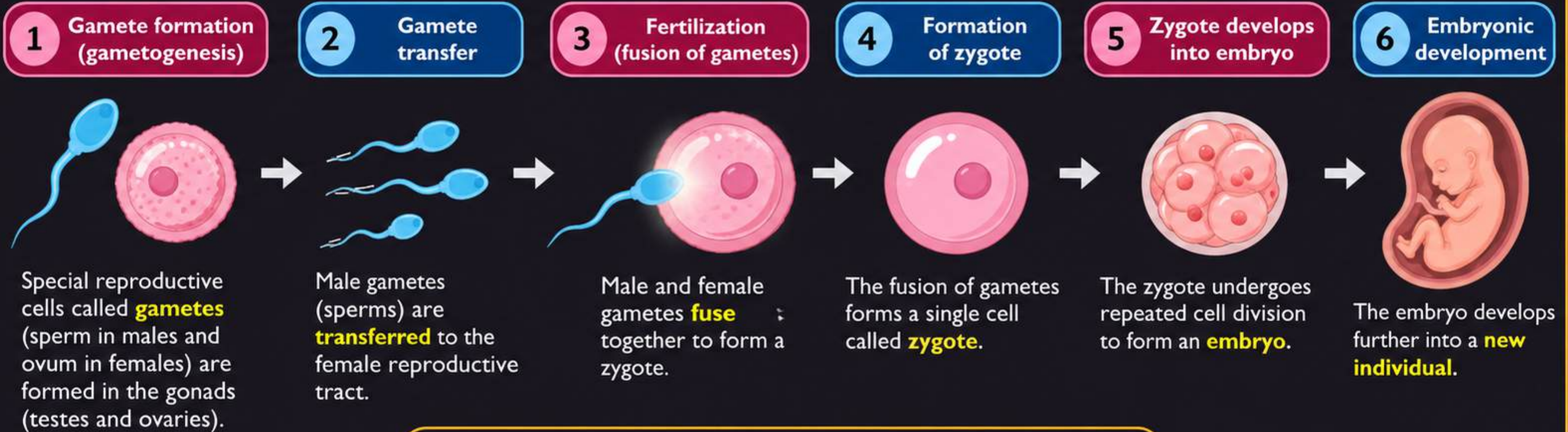
Advantages of sexual reproduction

- Variation in offspring – Sexual reproduction produces variations in the population.
- Better adaptability – Variations help organisms adapt to changing environment.
- Evolution – It promotes evolution by natural selection.
- Survival of species – Increases chances of survival of the species in adverse conditions.

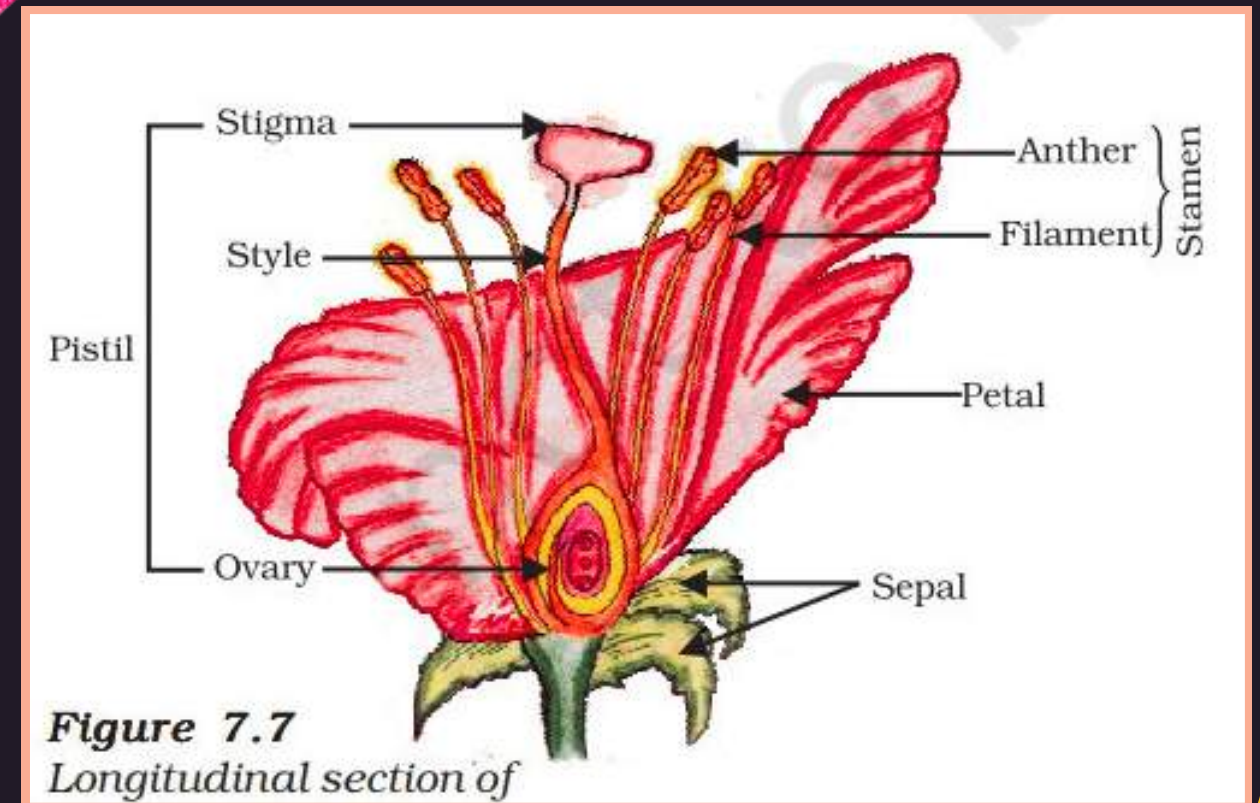
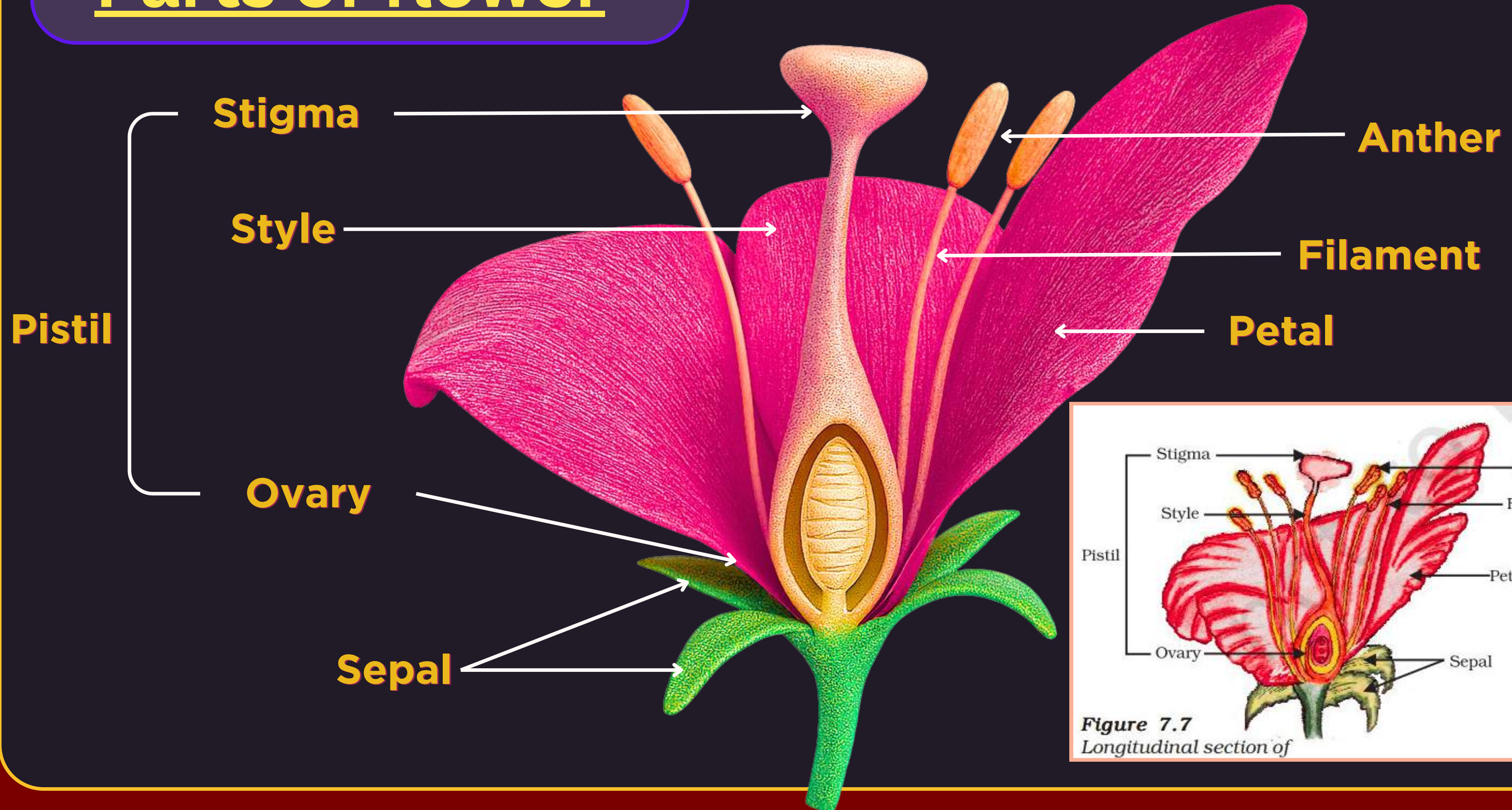


Steps in Sexual Reproduction

These steps occur both in plants as well as animals.



Parts of flower



Parts of flower and their functions

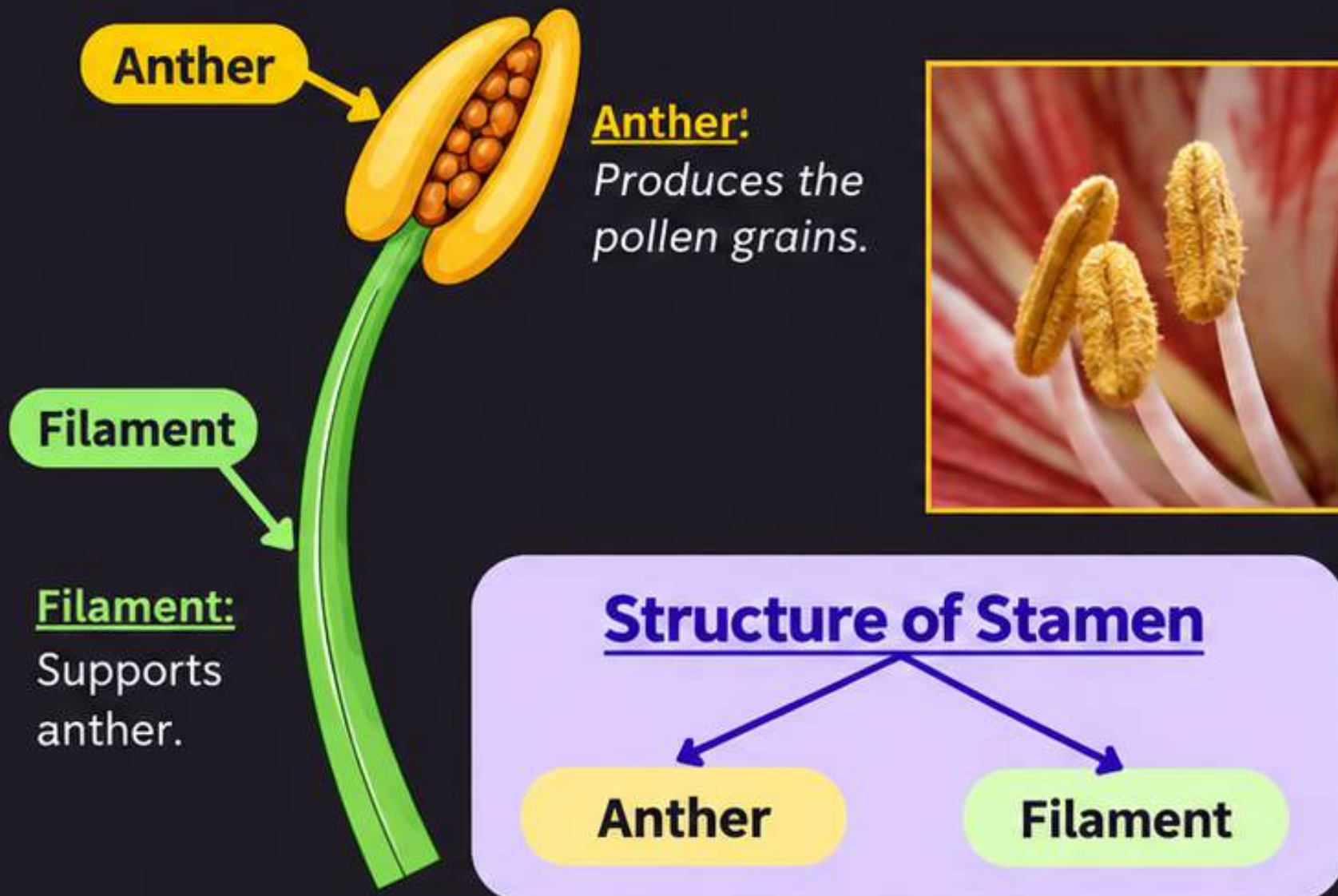
Part	Function
Sepals	Usually green; protect the flower in the bud stage .
Petals	Brightly coloured and sometimes fragrant; attract insects/pollinators .
Anther	Produces pollen grains , which contain male gametes .
Filament	A stalk that supports the anther . Together with anther, it makes the stamen (male part) .
Stigma	Sticky tip of pistil; receives pollen grains during pollination .
Stamen	Elongated tube that connects stigma and ovary ; pollen tube grows through it to carry male gametes.
Ovary	Swollen basal part of pistil; contains ovules ; after fertilisation it develops into a fruit .
Ovule	Present inside ovary; contains the female gamete (egg cell) ; site of fertilisation ; develops into a seed .

- **Pistil:** The **female reproductive part** of a flower. Present at the centre of the flower.
- **Stamen:** The **male reproductive part** of a flower.

Reproductive structures in flowers

Male reproductive system / Androecium

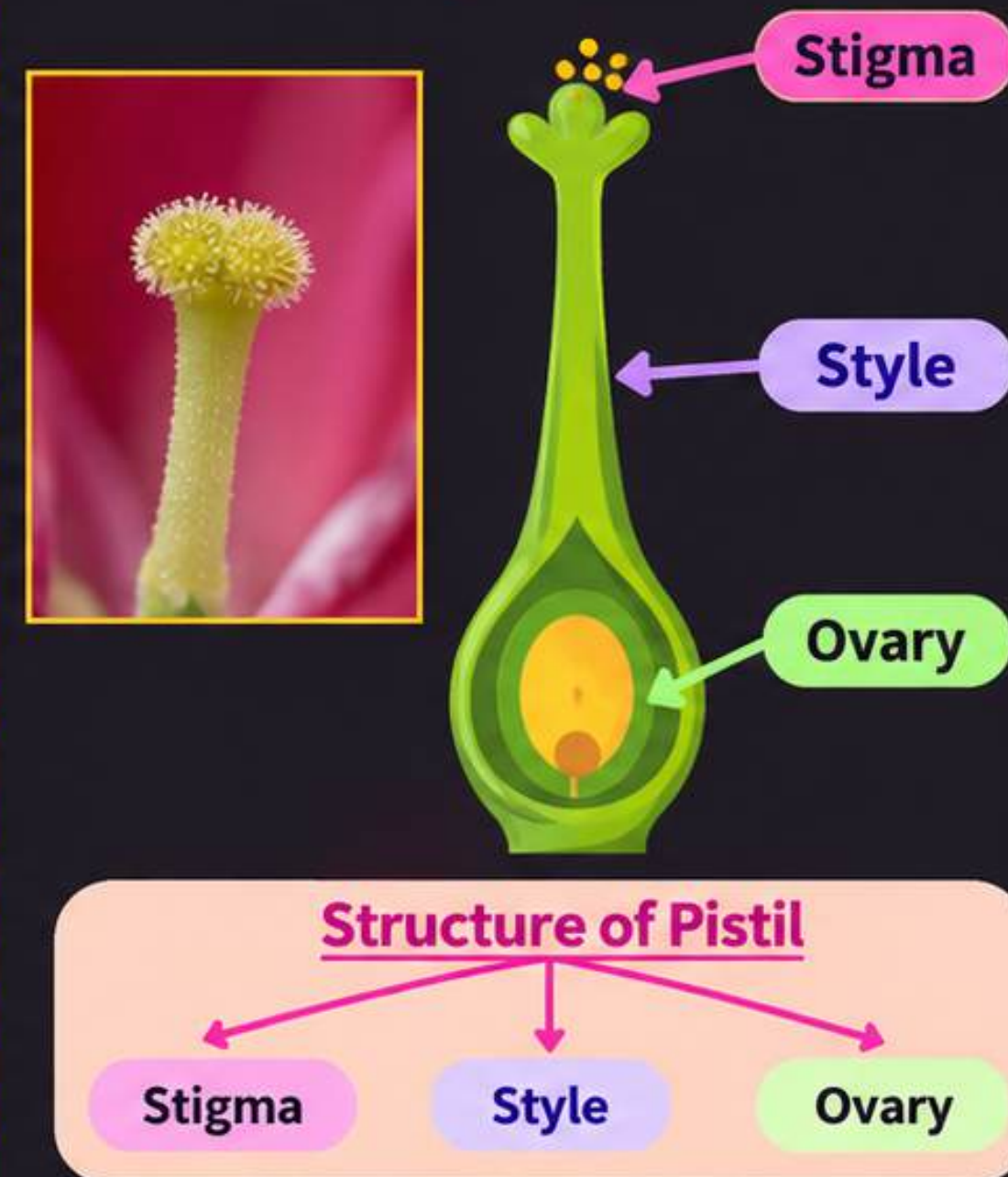
Androecium consists of stamens.



- Male gamete is present inside pollen grain.

Female reproductive system / Gynoecium

Female reproductive system consists of pistil.



Stigma:

It is the terminal part of the carpel, which may be sticky. It helps in receiving the pollen grains during pollination.

Style:

It is the middle elongated part of the carpel. It helps in the attachment of stigma to the ovary.

Ovary:

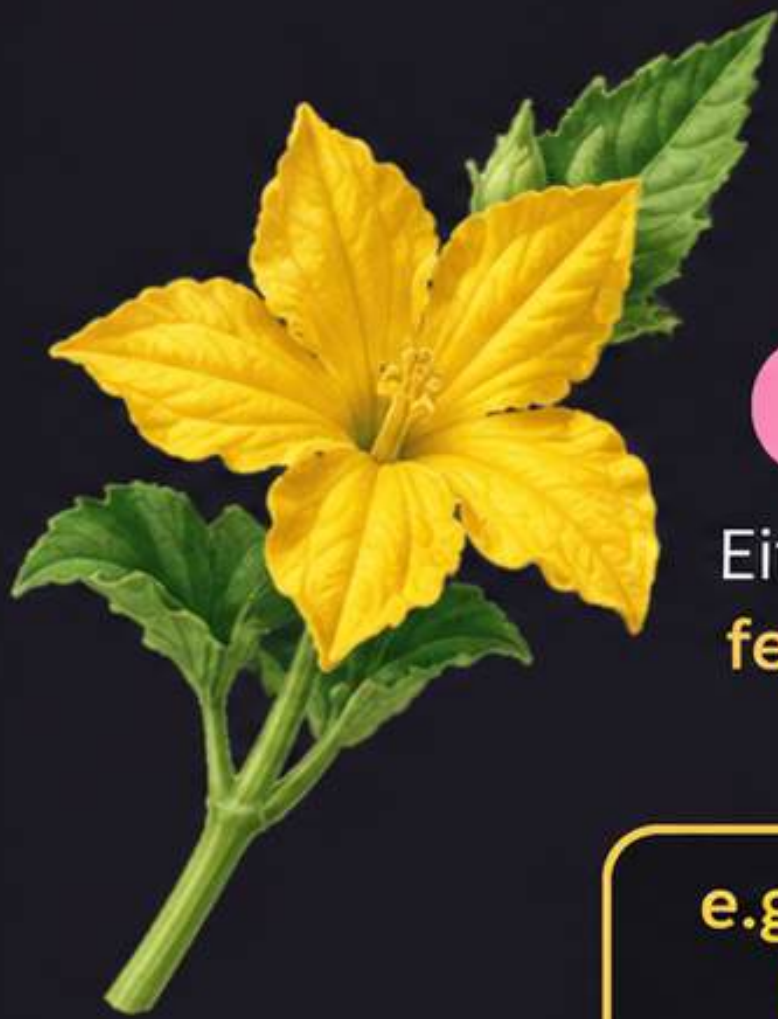
It is the swollen bottom part of the carpel. It contains ovules having an egg cell (female gamete).

In flowering plants, stamen is the male reproductive part and pistil is the female reproductive part.

Sexual Reproduction in Flowering Plants

In flowering plants, sexual reproduction occurs through the fusion of **male gametes (pollen)** and **female gametes (egg)** inside the ovule. Pollination transfers pollen to stigma, followed by fertilization. The ovule develops into a seed and the ovary into a fruit. All flowering plants belong to **Angiosperms**.

Types of flowers



Unisexual:

Either **male (stamen)** or **female (pistil)** parent is present.

e.g. Papaya, Watermelon, Coconut, Cucumber

Bisexual:

Both **male (stamen)** and **female (pistil)** parent is present.

e.g. Hibiscus, Mustard, Rose, Pea



Check yourself

Q. Part(s) of a flower which attracts insects for pollination is (are)

- (a) Petals and Sepals**
- (b) Anther and Stigma**
- (c) Petals only**
- (d) Sepals only**

Check yourself

Question: A student observes a flower before and after fertilization. After some time, the flower develops into a fruit containing seeds.

- 1. What is formed when the male and the female gamete cells fuse.**
- 2. What does the zygote develop into?**
- 3. What happens to ovule after fertilization?**
- 4. What happens to the ovary?**
- 5. How can the seed ultimately give rise to a new plant?**

Pollination

The process of transfer of pollen grains from the anther (male part) to the stigma (female part) of a flower.

Types of Pollination

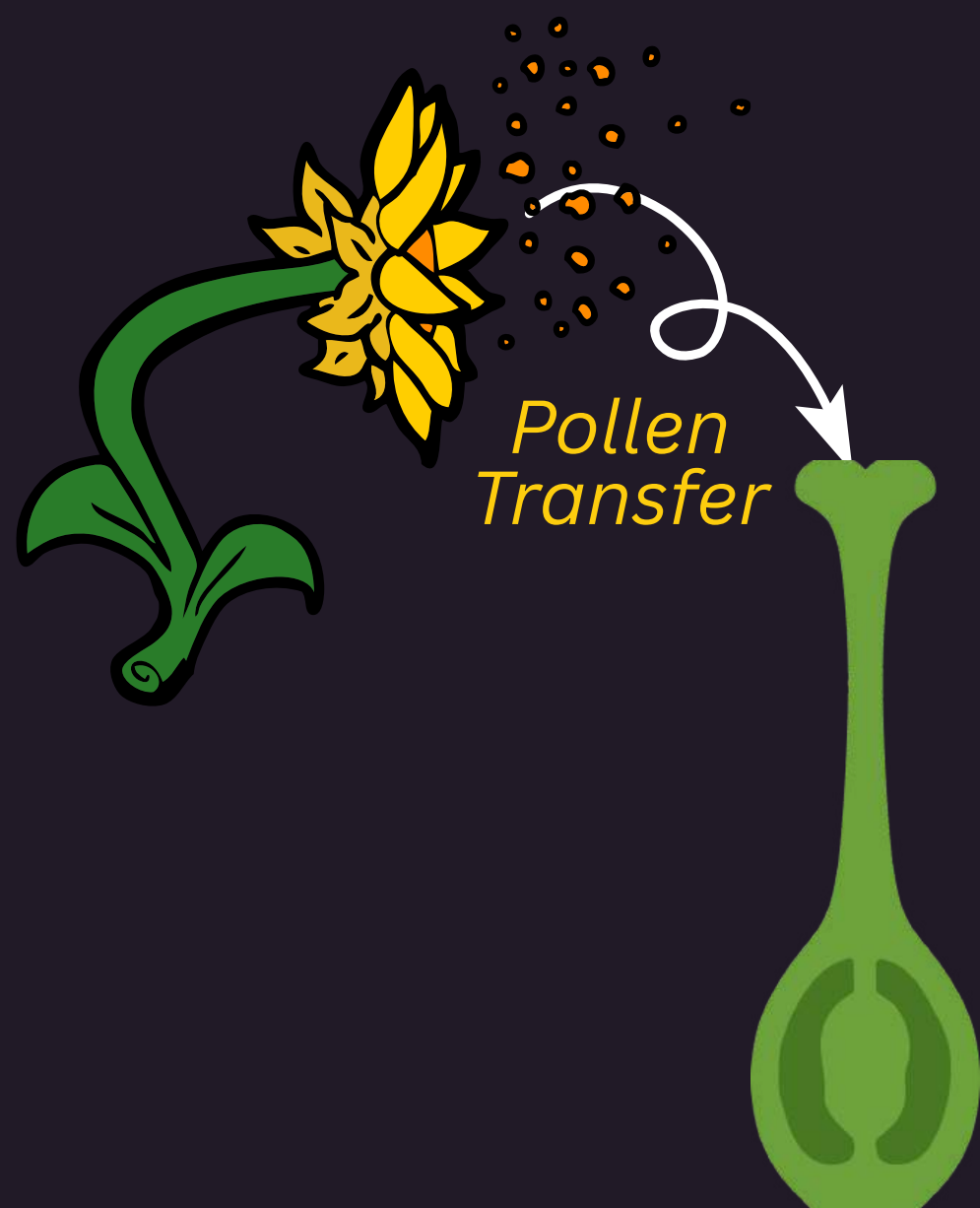
Self Pollination

Pollen transfer occurs from anther to stigma of same flower
e.g.- Mustard

Cross Pollination

Pollen transfer occurs from anther to stigma of different flower
e.g.- Papaya

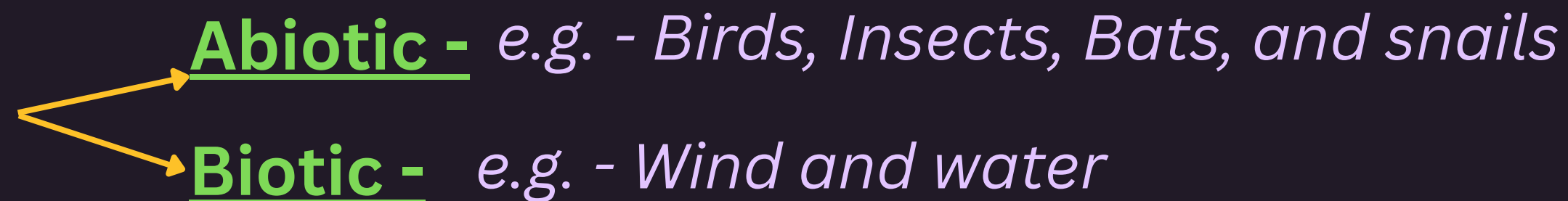
After pollination fertilization occurs.



Differentiate between self-pollination and cross-pollination.

Self-Pollination	Cross-Pollination
Pollen grains are transferred from the anther to the stigma of the same flower .	Pollen grains are transferred from the anther of one flower to the stigma of another flower of the same species.
Occurs in bisexual flowers (e.g., Hibiscus, Mustard).	Occurs in unisexual or bisexual flowers (e.g., Papaya, Watermelon).
No external agents required.	Requires agents like wind, water, or animals (insects, birds, etc.).
Offspring show less variation (genetically similar to parent).	Offspring show more variation (due to mixing of gametes from two plants).

Types of Pollinator



Check yourself

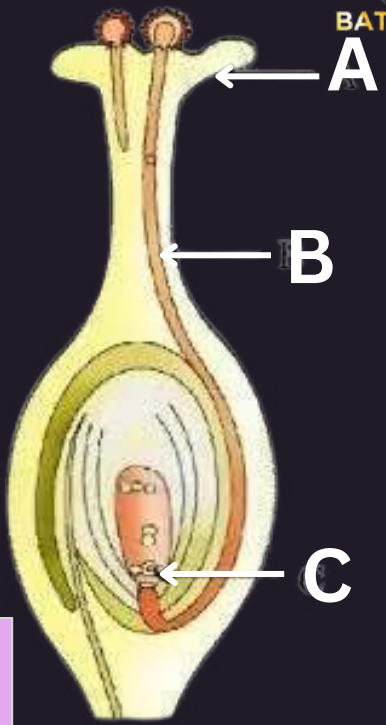
QUES: A flower receives pollen from another flower of the same plant.
This is the best example of

- a) Self-pollination
- b) cross-pollination
- c) Fertilization
- d) germination

2015, 2017, 2019, 2021, 2023, 2025

Q. (i) Differentiate between self-pollination and cross-pollination. Write two advantages of cross-pollination.

(ii) Identify A, B, and C in the diagram given below and write one function of each.



Self-Pollination	Cross-Pollination
Transfer of pollen grains from anther to stigma of <i>same flower or same plant</i> .	Transfer of pollen grains from anther of <i>one flower</i> to stigma of <i>another flower on a different plant</i> .
Does not require agents (like wind/insects).	Requires pollinating agents (like wind, water, or insects).
No variation in offspring.	Produces variations in offspring.

- A. Stigma:** Receives pollen grains during pollination.
- B. Pollen tube:** Carries male gametes from the pollen grain to the ovule.
- C. Female germ-cell / Egg cell:** Fuses with the male gamete to form a zygote.

2015, 2019, 2022

Q. Assertion and Reasons

Assertion (A): Cross-pollination leads to greater variations than self-pollination.

Reason (R): In cross-pollination, pollen grains come from a genetically different plant.

1. Both Assertion and Reason are true
2. Assertion and Reason are true and Reason is the correct explanation of assertion
3. Assertion is false Reason is true
4. Assertion is true Reason is false

Check yourself

Ques: Assertion: A flower undergoes cross-pollination without the pollen being transferred to its stigma.

Reason: Cross-pollination involves the transfer of pollen from one flower to another.

- a) Both A and R are true, and R is the correct explanation of A.**
- b) Both A and R are true, and R is not the correct explanation of A.**
- c) A is false and R is true.**
- d) R is false but A is true.**

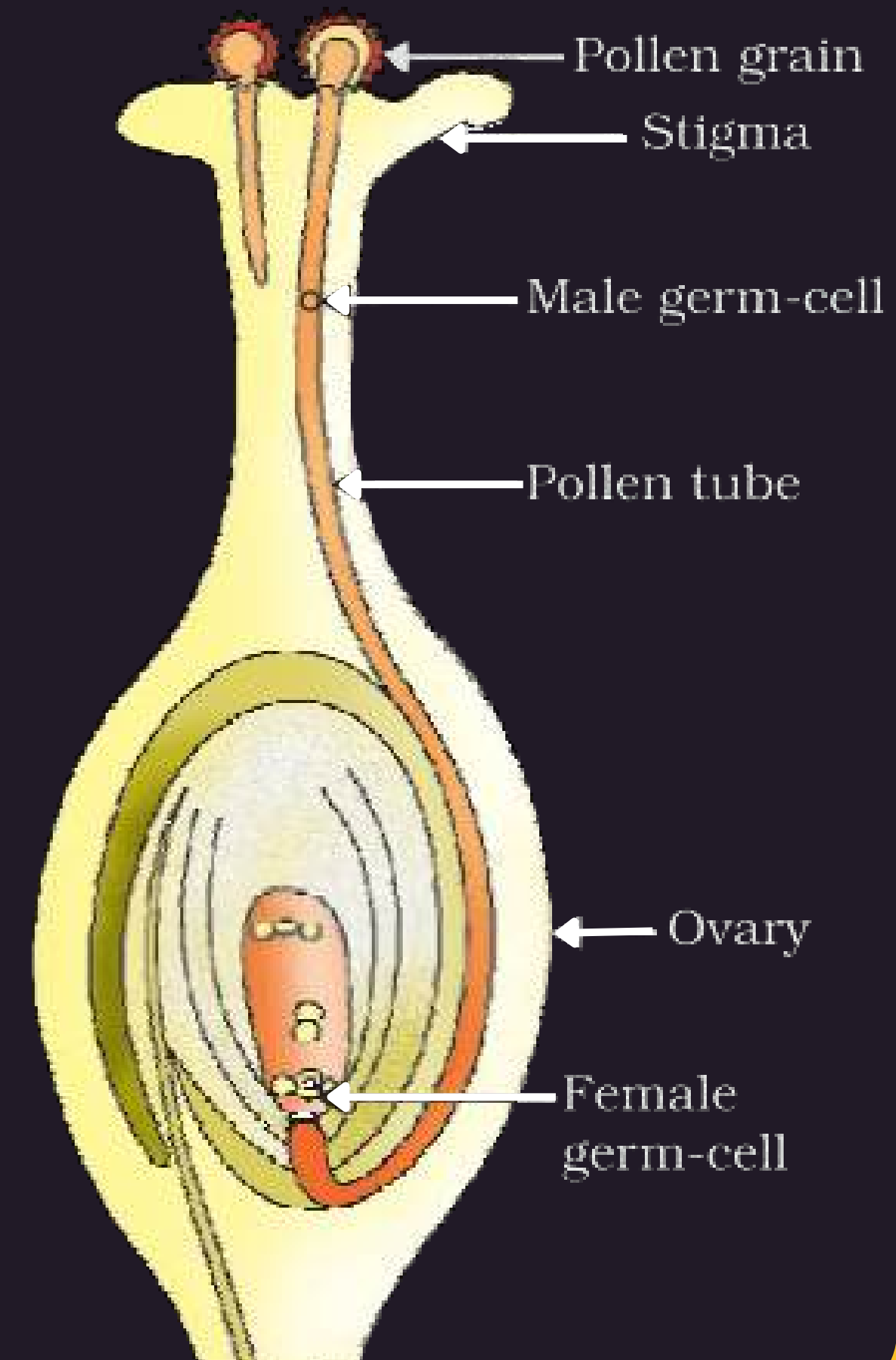


FERTILIZATION

Fertilisation is the process of fusion of male gamete (pollen grain nucleus) with the female gamete (egg cell) inside the ovule.

Process:

1. After pollination, pollen grain lands on the stigma.
2. Pollen grain germinates and produces a pollen tube.
3. The pollen tube grows through the style and reaches the ovary.
4. Male gamete travels through the tube → fuses with the egg cell in the ovule.
5. This fusion forms a zygote.

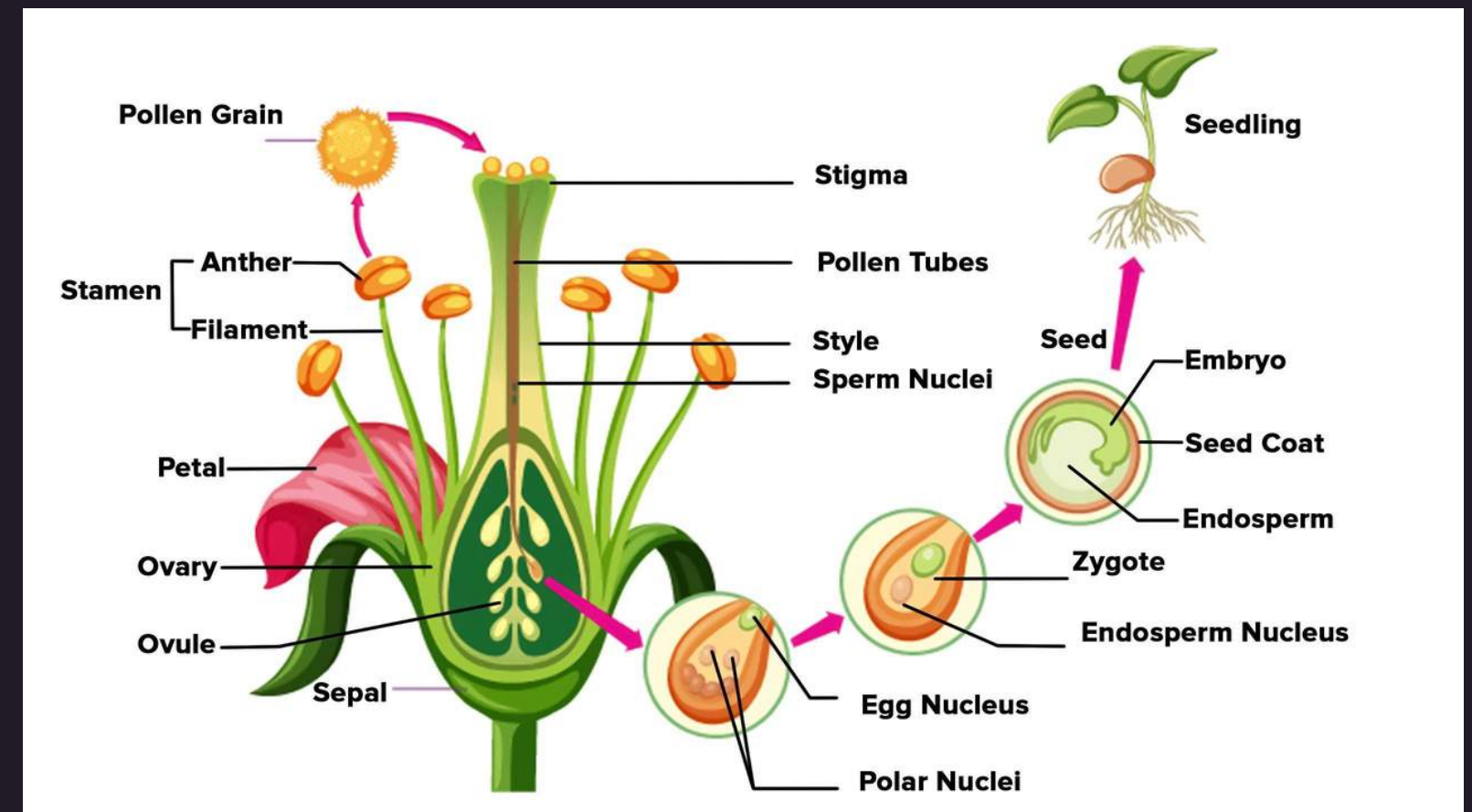


Post - Fertilization

Post-fertilisation refers to all the changes that occur in a flower after fertilisation to form seeds and fruits.

Changes After Fertilisation:

1. **Zygote** develops into an embryo.
2. **Ovule** becomes a seed (with protective coat).
3. **Ovary** develops into a fruit.
4. **Petals, sepals, stamens, style, stigma** usually wither and fall off.



Zygote → **Embryo**

Ovule → **Seed**

Ovary → **Fruit**

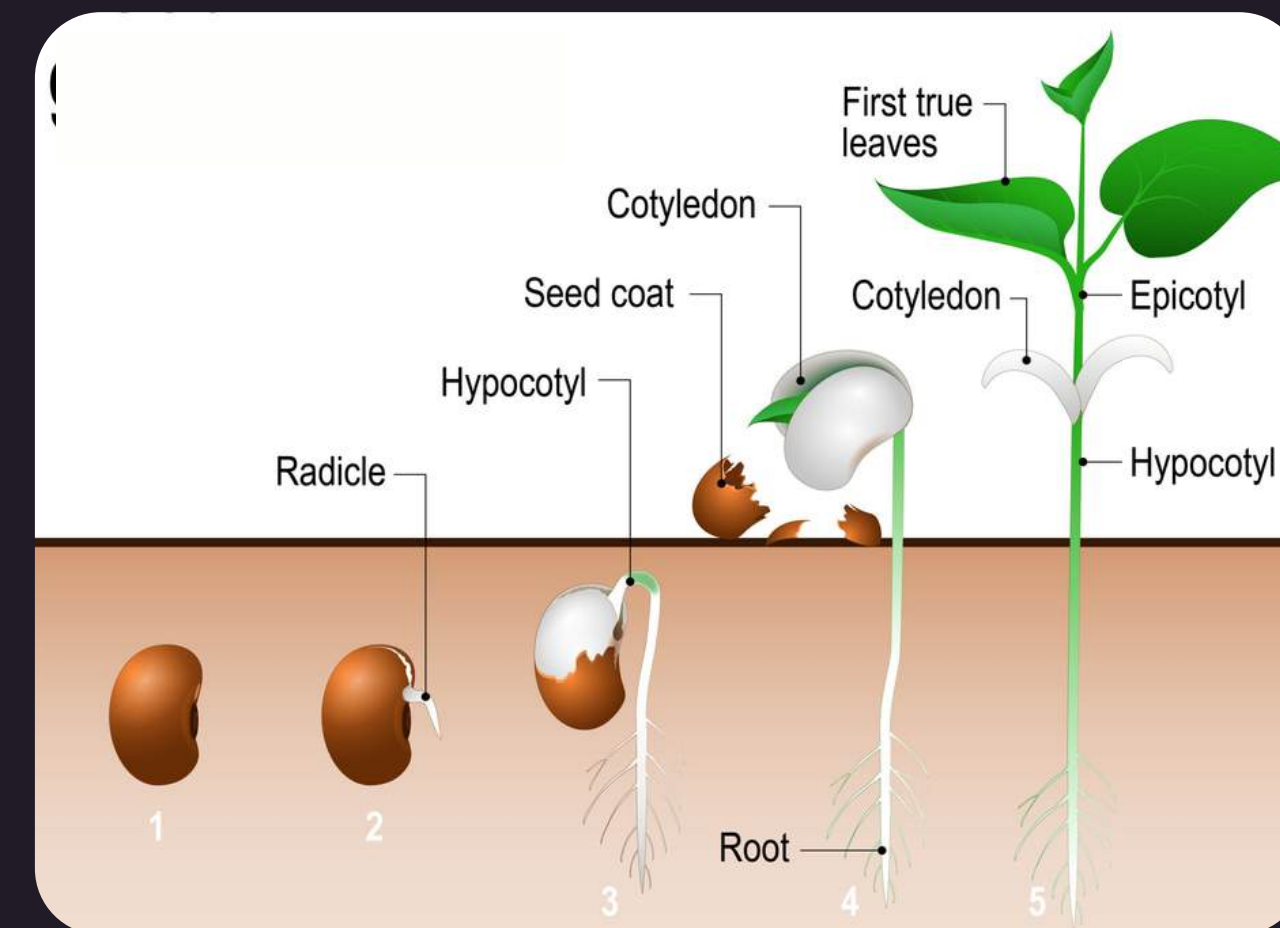
SEED-GERMINATION

Seed is the final product of Sexual reproduction.

- **Germination** is the process by which a seed develops into a new plant (seedling) under suitable conditions.

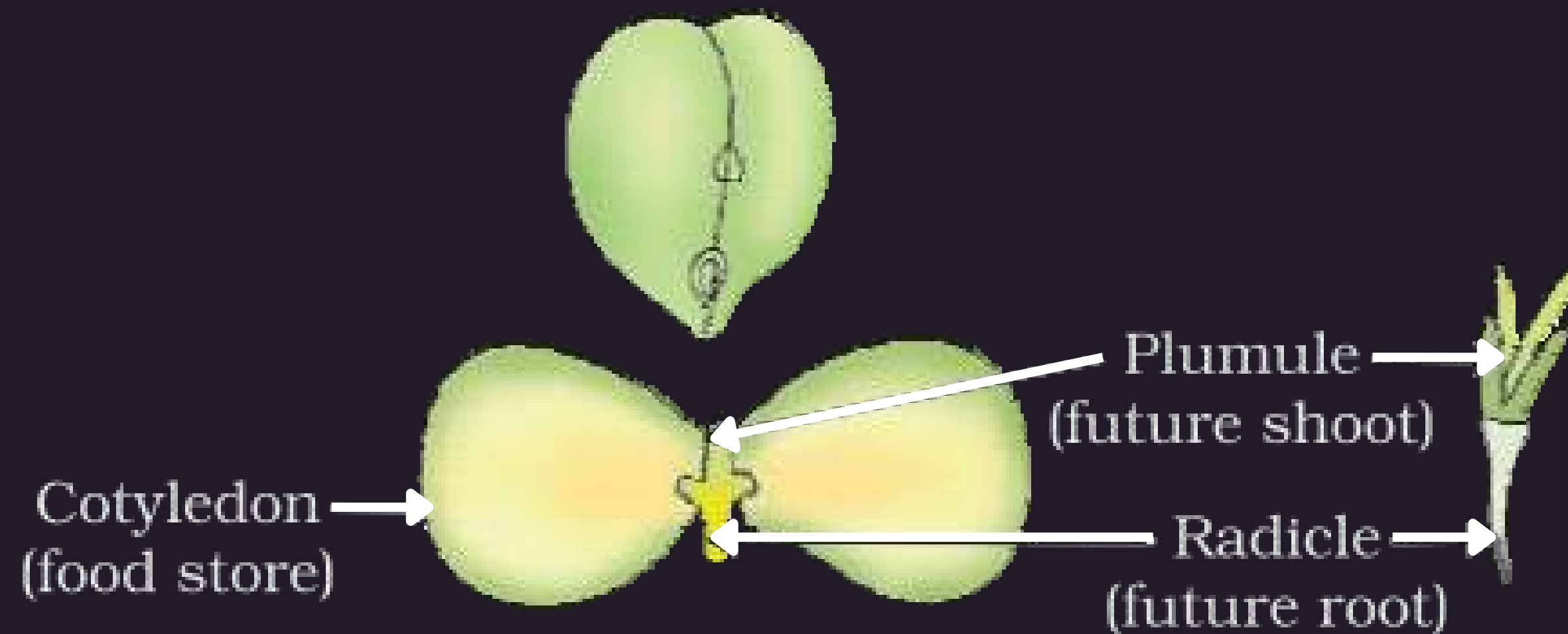
PARTS OF A SEED:

- **Seed Coat:** The outer protective covering of the seed. Protects the seed and embryo from mechanical injury, drying, and infection.
- **Cotyledons:** Also called seed leaves. Contain stored food that nourishes the embryo during germination.
- **Embryo (Baby Plant):** The future plant present inside the seed.



Made up of two main parts:

- **Radicle** → develops into the root system.
- **Plumule** → develops into the shoot system.



2016, 2017, 2019, 2020, 2023, 2025

Q. What happens to the following parts of a flower after fertilisation? (i) Ovary (ii) Ovule (iii) Petals (iv) Stamens

Part of Flower	What Happens After Fertilisation
(i) Ovary	Develops into a fruit .
(ii) Ovule	Develops into a seed .
(iii) Petals	Wither and fall off.
(iv) Stamens	Dry up and fall off.

Check yourself

Ques: After fertilization, a flower undergoes several changes. Which combination is correct?

- a) Ovule → fruit, ovary → seed**
- b) Ovule → seed, ovary → fruit**
- c) Ovule → embryo, ovary → pollen**
- d) Ovule → fruit, ovary → pollen**

Check yourself

Ques: How is the process of pollination different from fertilization? [NCERT]

Pollination	Fertilisation
Transfer of pollen grains from the anther to the stigma of a flower.	Fusion of male gamete (pollen nucleus) with the female gamete (egg cell) inside the ovule.
It is a physical process (movement of pollen).	It is a biological process (fusion of gametes).
Can be of two types → Self-pollination and Cross-pollination .	Only one type – union of male and female gametes.
Occurs before fertilisation .	Occurs after pollination .
No zygote or seed is formed.	Results in zygote formation , leading to seed and fruit .

Check yourself

Q. What happens to the following after fertilisation:

- (a) zygote
- (b) ovule
- (c) ovary
- (d) petals and stamens?

- (a) Zygote → develops into an embryo.
- (b) Ovule → develops into a seed.
- (c) Ovary → develops into a fruit.
- (d) Petals and stamens → wither/dry up and usually fall off.

Check yourself

Q. Why is the stigma generally sticky? What happens after a compatible pollen grain lands on it?

Answer: The stigma is generally sticky so that it can receive and hold pollen grains during pollination.

After a compatible pollen grain lands on the stigma, it germinates and forms a pollen tube. The pollen tube grows through the style towards the ovary and carries the male gamete to the ovule, where fertilisation occurs.

Check yourself

Q. Two identical flowers P and Q are taken. In P, the stigma is covered before pollen grains reach it. Q is left uncovered. Which flower is more likely to form seeds? Explain.

Answer: Flower Q is more likely to form seeds.

In flower P, the stigma is covered, so pollen grains cannot reach it and pollination does not occur. Therefore, fertilisation cannot take place and seeds will not form.

In flower Q, the stigma is exposed, so pollen grains can land on it. The pollen grain can then germinate, form a pollen tube, and carry the male gamete to the ovule for fertilisation, leading to seed formation.

Check yourself

Ques: A flowering plant has two flowers, A and B. Pollen grains from the anther of flower A are transferred to the stigma of flower B. In another situation, the pollen from the anther reaches the stigma of the same flower.

1. Name the type of pollination in both flowers.
2. Mention two agents that can bring about the transfer of pollen between flowers?
3. Explain why pollen must reach stigma before fertilization can occur
4. What structure develops from pollen grain to carry the male germ cell towards the female germ cell?

1. Flower A → Flower B: Cross-pollination

Same flower: Self-pollination

2. **Two agents of pollination: Wind and Insects**
3. **Pollen must reach the stigma because after landing on the stigma, the pollen grain germinates and forms a pollen tube. This tube carries the male gamete towards the ovule, where fertilisation occurs.**
4. **The structure is the pollen tube. It develops from the pollen grain and carries the male germ cell towards the female germ cell.**

Check yourself

Q. Three similar flowers are subjected to the following treatments: in flower A, stamens are removed before the anthers mature; in flower B, the stigma is removed before pollination; and in flower C, the ovary is removed after pollination but before fertilisation. Predict how each treatment would affect pollination, fertilisation and the formation of seeds/fruits. Give reasons for your answer.

- **Flower A — Stamens removed before anthers mature:**

Self-pollination will be prevented because the flower cannot produce its own pollen. However, cross-pollination can still occur if pollen from another flower reaches its stigma. If fertilisation occurs, seeds and fruit can still form. This follows from the lecture's distinction between self- and cross-pollination.

- **Flower B — Stigma removed before pollination:**

Pollination will not occur because the stigma is the part that receives pollen grains. Therefore, fertilisation will not take place and seeds and fruit will not form.

- **Flower C — Ovary removed after pollination but before fertilisation:**

Fertilisation cannot occur because the ovules containing the female gamete are present inside the ovary. Hence, seeds will not form. Also, since the ovary develops into the fruit after fertilisation, fruit formation will also not occur.

Check yourself

Ques: Assertion: The formation of a seed does not immediately mean that a new plant has developed.

Reason: The embryo present inside the seed develops into a seedling under appropriate conditions through germination.

- a) Both A and R are true, and R is the correct explanation of A.
- b) Both A and R are true, and R is not the correct explanation of A.
- c) A is false and R is true.
- d) R is false but A is true.

Check yourself

A researcher removes the stigma of a flower before pollen can reach it.

Which event would be directly prevented?

- a) Formation of ovule**
- b) production of pollen grains**
- c) transfer of pollen to female reproductive part**
- d) Formation of fruits after successfully fertilization**

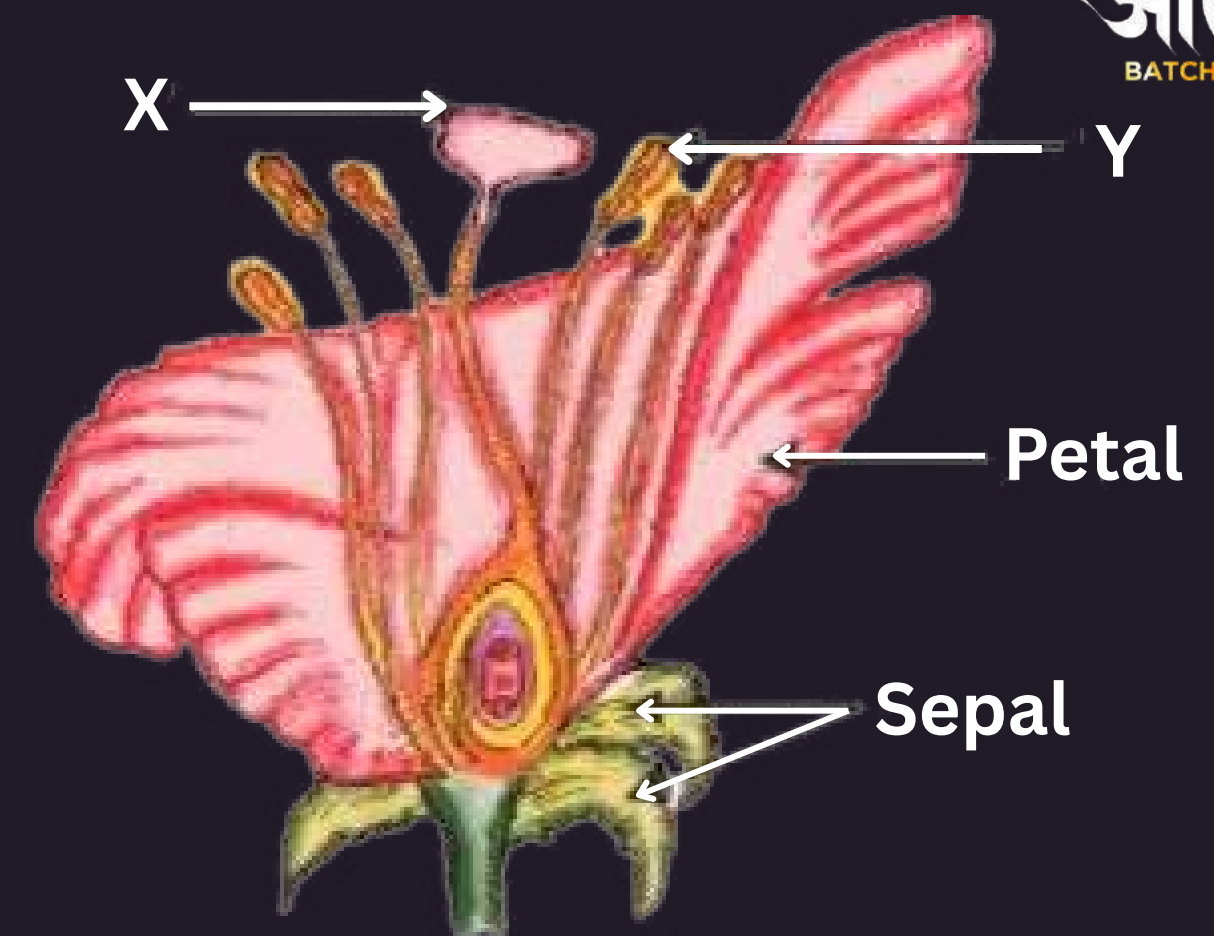
2015, 2017, 2019, 2021, 2023, 2025

Q.(i) Identify the parts 'X' and 'Y' in the figure given below:

(ii) Name the yellowish coloured structures produced by the part labelled as 'Y'.

(iii) Write the name of the process by which these are transferred to the part labelled as 'X'.

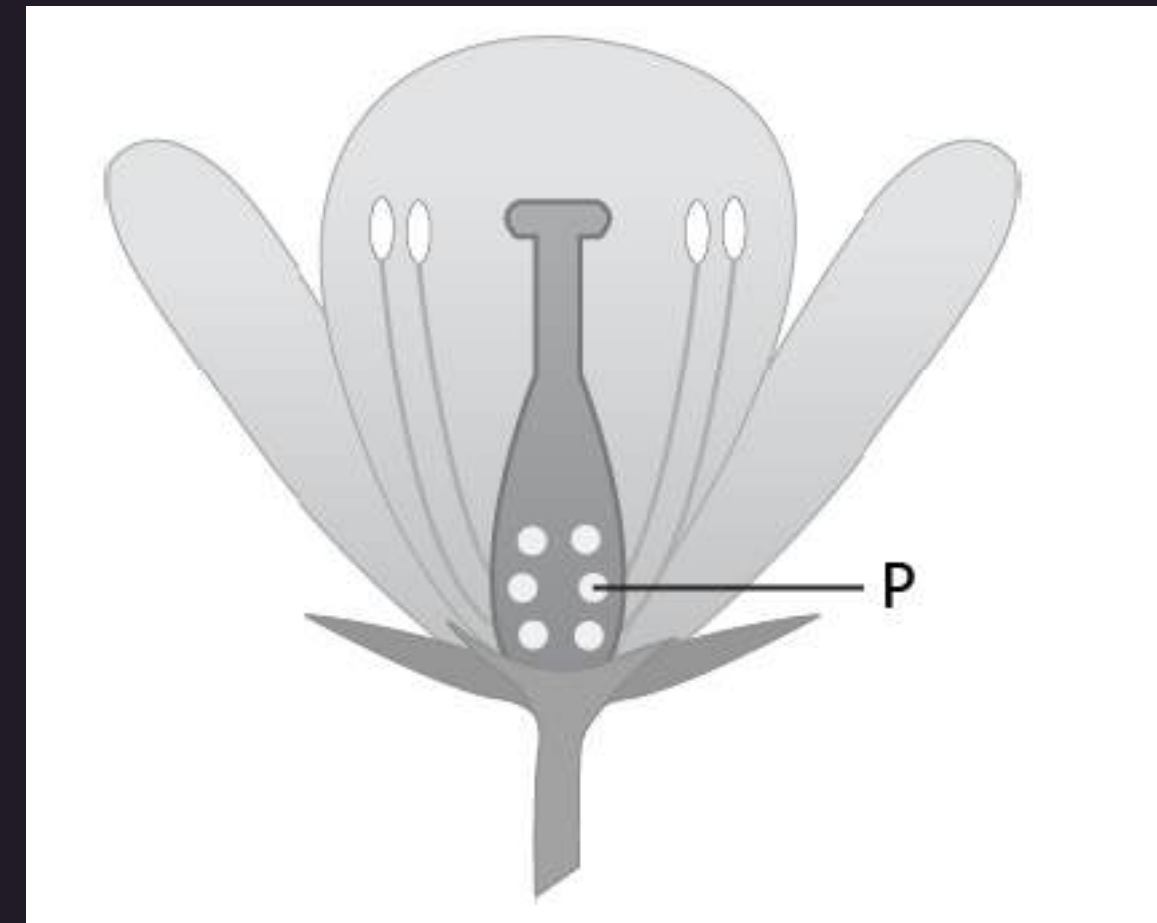
(iv) Explain the process of seed formation in a flowering plant.



Check yourself

Q. The image shows the structure of a flower. Which of the following processes will likely be disturbed or not occur if the labelled part is removed from the flower?

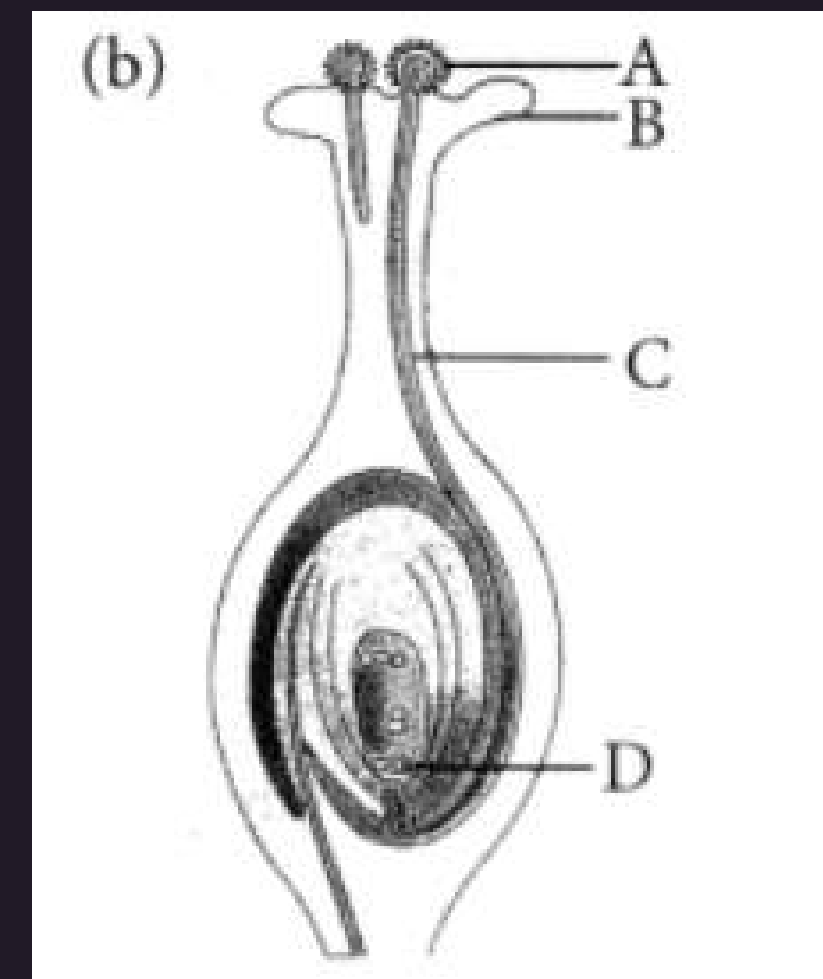
- a) Formation of fruit**
- b) Transport of pollen**
- c) Formation of pollen**
- d) Development of the pollen tube**



Check yourself

- Q. (i) Name the part marked A in the diagram.
 (ii) How does A reaches part B?
 (iii) State the importance of the part C.
 (iv) What happens to the part marked D after fertilisation is over?

- (i) Part A is the Pollen grain.
 (ii) A reaches part B by the process of pollination. Pollen grains are transferred from the anther to the stigma (part B) by agents like wind, water or insects.
 (iii) Part C is the style.
 Its importance: It provides a passage for the pollen tube to grow from the stigma to the ovary for fertilisation.
 (iv) Part D is the ovule.
 After fertilisation, the ovule develops into a seed.





Ye lo
Phool

